# Working With Fire Departments to Adapt and Implement Evidence-Based Programs That Increase Uptake of Smoke Alarms: A Case-Series Report

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**We describe a partnership between an academic injury center and three U.S. fire departments to adapt and implement strategies for promoting smoke alarm programs. Each fire department identified the aims and parameters for a new promotion campaign for their smoke alarm programs. Promotion was directed toward residents in each department's catchment area who were eligible for the smoke alarm program. All three departments independently elected to use an automated telephone message to promote their smoke alarm programs. Uptake of smoke alarm installation services ranged between 0.02% and 2% of the calls placed. In Rochester, automated calls were sent to all residential landlines via the city's nonemergency call center; requests for smoke alarms increased by a factor of 7.5 in the month following the campaign. In Grand Rapids, automated calls were sent to 6% of the households served due to the telecommunications infrastructure; because of the limited reach, the overall number of requests remained unchanged, and the number of callers citing the automated calls was less than the number of requests callers reported were motivated by Grand Rapids' existing promotion strategies. In Cloquet, the automated calls were broadcast on a rolling basis to geographic clusters of households; although the number of requests remained unchanged, fire district officials were pleased to reduce driving time between appointments which allowed volunteers to offer more home visit appointments. Automated telephone calls can be a valuable tool for promoting smoke alarm programs, but fire departments should carefully identify how dissemination strategies can best complement existing program efforts.**

Residential fires are a significant public health issue, affecting approximately 380,000 U.S. households per year and resulting in over 2700 deaths and 11,000 injuries.[1] There is robust evidence establishing smoke alarms as effective in reducing the risk of death when a residential fire occurs,[2,3] and an instructive evaluation literature of programs designed to increase smoke alarm prevalence.[4–10] However, evidence about smoke alarms and the programs that support their use is insufficient. Homes without working smoke alarms are consistently documented, especially in high-risk settings,[11–16] leaving too many vulnerable to preventable death when a fire breaks out in their home.

The most impactful smoke alarm programs (as measured by homes equipped with a working smoke alarm) are delivered by the fire service and offer home visits with education, a home inspection as well as free or low-cost alarms and installation.[7] The benefit of neighborhood canvasing to provide installation services (as opposed to free alarms without installation) is a consistent finding.[4,10] Direct installation has been evaluated as part of door-to-door canvasing events,[5] partnering with other home visiting services,[17] and by responding to resident requests for a smoke alarm.[18] How to disseminate information about smoke alarm installation programs to maximize uptake by community members has been the subject of less scholarship.

A partnership between members of this study team and the Baltimore City Fire Department (BCFD) compared two methods for promoting the BCFD smoke alarm installation program—disseminating an automated message through the City's 311-call system and purchasing a Facebook advertisement that targeted City residents.[18] Callers requesting smoke alarm installations following the automated calls and the Facebook ads most often cited the automated message from the BCFD Fire Chief as the reason for their smoke alarm installation request (n = 458) compared to a Facebook ad (n = 25).[18] The automated system offered a cost-effective promotional strategy for increasing requests for the City's smoke alarm installation program.

Many fire departments have active smoke alarm programs to increase the prevalence of homes with working smoke alarms in the communities they serve. However, the impact of these programs relies on adapting the evidence-based intervention to fit their locality's needs, and resources and capacity for implementing said programs. Recognition that a one-size-fits-all intervention approach likely fits few is increasingly prominent in the implementation literature.[19,20] The importance of adaptation and fidelity when scaling up evidence-based interventions is critical to realizing the effects demonstrated

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in effectiveness studies. How to balance fidelity to an intervention with adapting that intervention to account for local context is a question for which there is some guidance.[19,20]

Here, we report on a project with the goal of partnering an academic injury center with fire departments to adapt and implement strategies for promoting existing smoke alarm programs in order to increase uptake of those programs. We describe the resulting campaigns and assess the impact of the adapted campaign approaches within and across the participating departments for the purpose of informing future efforts to increase uptake of fire department smoke alarm installation programs.

## METHODS

The academic injury center faculty (Center Faculty) worked with fire departments in Rochester, NY, Grand Rapids, MI, and Cloquet, MN (Table 1) to identify and implement strategies to promote each jurisdiction's smoke alarm program. A fourth city was initially included in our sample but was unable to complete the project due to personnel changes. Center Faculty invited these cities to participate because they had active smoke alarm programs and a program coordinator who was enthusiastic about trying a new approach to promote their smoke alarm program. The work was funded by a Fire Prevention and Safety (FP&S) grant through FEMA's Assistance to Firefighters Grant program. The FP&S grants are open to fire departments and other organizations engaged in fire prevention and community risk reduction to help cover the costs of materials and labor. Each department was asked to review their smoke alarm program to determine the capacity of their program for home visits and identify where there may be opportunity for improvement. Departments listed previous promotion efforts, routinely collected program metrics, and available resources.

The research team reviewed the information provided by the departments and offered to each feedback about target populations and new promotion methods to increase uptake of their smoke alarm programs. Through these discussions, the departments defined the components and parameters for a new promotion campaign for their smoke alarm programs (Figure 1). Each department identified a set of measures to assess the impact of their campaign. As part of their participation, each department received smoke alarms for distribution and was offered $500 to cover other expenses related to implementing the campaign. Funds from the project grant paid for the smoke alarms and the cash support. The methods of this project were reviewed and approved by the Johns Hopkins Bloomberg School of Public Health Institutional Review Board.

Each department provided the research team with data about the number of requests for home visits, and how callers heard about the smoke alarm program for a period at least 3 months before they started the campaign through 3 months following implementation of the campaign. Additional data, such as the number of smoke alarms installed, were available for those departments that routinely collected those measures prior to the project. We cleaned and coded these data using SAS 9.3 (SAS Institute, Cary, NC) and tabulated the cleaned dataset to assess whether the new promotional campaign strategies were associated with uptake of the smoke alarm programs in each jurisdiction.

At the conclusion of the campaign the team visited each of the sites, and presented the findings about each community's effort to fire department leadership and other stakeholders. During these meetings, the research team solicited feedback from fire department team members and debriefed the team about implementation processes. Additional feedback is reflected through coauthorship of fire department team members.

## RESULTS

The three participating fire departments vary in terms of the size of the populations they serve, although all three smoke alarm installation programs reach between 0.6% and 0.9% of the homes in their communities each year (Table 1). Each department articulated the goal of the promotional campaign reported herein as either to increase the number of homes included (Rochester) or streamline their current programw efforts (Grand Rapids and Cloquet) and used automated calls and/or postcards to achieve their goal.

**Table 1.** Participating fire departments

| Fire Department | Rochester Fire Department, NY | Grand Rapids Fire Department, MI | Cloquet Area Fire District, MN |
|---|---|---|---|
| Background | | | |
| Population served | 210,565[21] | 188,040[21] | 20,000 (est.) |
| Households served | 99,510 | 80,226 | 5989 |
| Average annual home visits conducted | 1800 | 1400 | 126 |
| **Intervention** | | | |
| Campaign Description | Automated Phone Call | Automated Phone Call | Automated Phone Call, Postcards |
| Households reached via promotional campaign | 59,000 | 5379 | 2409 |
| Home visits requested through campaign | 117 (0.2%) | 46 (0.9%) | 48 (2%) |
| Duration | 3 months | 4 months | 6 months |
| Total visits requested during intervention window | 206 | 407 | 146 |

**Promoting Smoke Alarm Installation Services: Planning A Promotional Campaign**

1. **Target Population**
Is there a target population within your service area that you would like to be the focus of your promotion activities? Examples of target populations include:
   - Children and older adults
   - race or ethnic groups
   - workers in a particular industry
   - people in a high-risk part of your service area

2. **Promotion Opportunities**
Given the available resources and your community's needs, which of these strategies may best complement your installation promotion efforts?
- **Canvassing.** Deploy teams of firefighters and/or volunteers to neighborhoods to install smoke alarms. The neighborhoods are selected based on risk, in response to a fire, or as part of a plan to cover a community. Canvassing may include providing advance notice to the neighborhoods that the teams will be visiting, and at homes with no response teams may leave materials that include instructions for how to request installation at another time.
- **Postcards**. Postcards are mailed to homes with information about the installation service.
- **Automated voice or text messages.** Voice or text messages using public or private systems deliver information about the installation services to land lines or cell phones can provide a mechanism for targeting specific populations at designated days and times.
- **Community Associations.** Using community associations as a forum for sharing information about installation services can occur through participation in meetings, and contributions to newsletters, listserves, NextDoor posts, and websites.
- **Other City Offices (e.g., mayor's office, police, housing, etc.).** City agencies can assist with promotion of installation services, and some (such as health and housing) may have a shared interest in the service.
- **Public service announcement.** Radio and/or television ads can be prepared for broadcast as a public service announcement.
- **Local news**. Outreach to newspapers, TV, or radio outlets to offer installation services as a news story can generate media attention to the installation program.
- **Social Media.** Paid advertisements that target a defined population or adding content about the installation program to an existing page are two strategies to reach Facebook users. Tweets that provide general information about installation services or those that are designed to capitalize on events or times when people may be more receptive to the message can be used to promote installation services.
- **Citizen Advocates.** Enlist private citizens and or local businesses to promote the campaign using incentives such as recognition with an award or a prize.
- **Billboard advertisements.** Electronic or standard billboards can provide a mechanism for promoting installation services.
- **Community service organizations**. Incorporating information about installation services or installation with service organizations can provide installation teams with access to high-risk, hard-to-reach people. A list of community service organizations in the four jurisdictions participating in this project is included below.
- **Industry.** Employers can provide a venue for providing information about smoke alarm installation services in a community. Time on meeting or training agendas to disseminate information about installation services, space in company newsletters, or during training sessions or information are all ways that partnerships with industry can help promote installation services.

3. **Measures**
Assessing the impact of different promotional strategies is important for informing future decisions about fire prevention resources. Following are questions we would like to have included.
- How did you hear about the smoke alarm installation program? (Allow for multiple responses.)
- How many children under 18 years live in the home?
- How many adults 65 and over live in the home?

4. **Community Partnerships**
Following are a list of community organizations that may provide opportunities for partnerships to promote smoke alarm installation services. [A tailored list of governmental and non-profit organizations providing services to at-risk families was provided to each participating department.]

**Figure 1.** Components and parameters for a smoke alarm promotional campaign.

## Rochester

Rochester is located in northwest New York with an estimated population of 210,000.[21] Rochester Fire Department (RFD) conducts approximately 1800 home visits to install smoke alarms per year. Residents can call the city's nonemergency 311 line and a fire suppression unit will respond within 2 hours to install smoke alarms free of charge. Prior to this campaign, RFD promoted their smoke alarm program through social media, the public information officer, on the RFD website and with door-to-door canvasing. RFD also checks and installs smoke alarms on EMS calls when possible and holds occasional canvasing events to install smoke alarms through door-to-door visits.

RFD sought to increase the number of home visit requests by bolstering its current promotion activities through this opportunity. After reviewing their smoke alarm program data and promotion efforts, discussing additional strategies with the research team, and speaking with their partners at the 311-call center, RFD developed and launched an automated phone message recorded by their fire marshal. Working with the 311-call center staff, they disseminated the 44 second message through the city's nonemergency 311-call system to approximately 59,000 phone numbers in November 2017, approximately 59% of the households in RFD's service area. The call center staff staggered calls between November 1 and November 30 to prevent overwhelming the fire department's capacity for home visits.

In the month following the start of the phone message campaign, requests for smoke alarms increased by a factor of 7.5 from 17 in October to 129 in November (Table 2). A total of 117 residents called in response to the automated message, yielding a return rate of less than 1% from the 59,000 calls placed. Response to the automated message was the most frequently cited reason for caller requesting a visit, with most callers responding during the month the calls were placed (Table 2). The promotion effort was well received by staff at the 311-call center who programmed the automated calls and responded to the influx of home visit requests. At the fire department, there was some initial concern among leadership about potential complaints from residents regarding automated calls and about overloading the responding suppression teams, however, they were pleased with the increase in smoke alarm requests and plan to continue using this method of promotion.

## Grand Rapids

Grand Rapids is located Western Michigan with a population of about 188,000. Grand Rapids Fire Department (GRFD) conducts approximately 1400 fire safety home visits and installs over 8500 smoke alarms each year. Residents call a central phone number to schedule an appointment for a home visit. Promotion and scheduling home visits are led by two full-time staff dedicated to this effort. GRFD advertised the smoke alarm program through canvasing, events, postcards, billboards, online, and though local partners.

GRFD sought to use this project as an opportunity to pilot a new promotion method in the hopes that it would streamline promotion, coordination, and installation efforts. The home visiting coordinators developed an automated phone message the fire chief recorded, and then they worked with

**Table 2.** Rochester: how did the caller requesting smoke alarm installation report hearing about the smoke alarm program?*

| | November 2017 | December 2017 | January 2018 |
|---|---|---|---|
| *Automated call* | 99 | 16 | *2* |
| Community association | 2 | 0 | 0 |
| Fire Dept. Personnel | 3 | 8 | 6 |
| Local news | 5 | 2 | 5 |
| PSA (TV or radio) | 1 | 2 | 8 |
| Social media | 0 | 2 | 0 |
| Word of mouth | 13 | 3 | 8 |
| Other | 6 | 5 | 10 |
| Total | 129 | 38 | 39 |

Italicized text indicates the new promotion strategy implemented.
*Shaded column indicates the month when automated calls were made.

the city's nonemergency 311-call system team to disseminate the message. Grand Rapids' 311-call system uses an analog phone system, and therefore the volume of incoming and outgoing calls is limited to the number of phone lines in the city's phone system. Consequently, circulation of the automated message was limited to certain days and hours when call volume was low so as not to exceed the capacity of the call center to make and receive calls needed for other city operations. A total of 5379 phone numbers, representing 6% of the households served, were submitted to the 311 system to be delivered over 14 days in October and November 2017.

Between October 2017 and January 2018, 46 residents requesting a smoke alarm home visit cited the automated call as the source of their information about the program (Table 3), resulting in less than 1% return on calls. This number was less than the number of requests callers reported were motivated by GRFD's existing promotion strategies; most calls were generated either by word of mouth (135 calls) or mailings (131 calls). Although GRFD leadership was supportive of this project, the city did not have the infrastructure needed to support and sustain the volume of automated calls to reach the city's population.

## Cloquet

Cloquet Area Fire District (CAFD) serves a rural population of about 20,000 people spread over 200 square miles on the eastern border of Minnesota. CAFD conducts about 120 home safety visits per year, generally on weekends, as determined by volunteers' availability. CAFD reaches approximately 2% of households in their service area each year. Prior to the campaign, CAFD promoted the program through their website, local news outlets, and social media. Residents call a central phone number to schedule an appointment for a home visit and the coordinator assigns the caller a slot on the volunteer calendar.

Because CAFD's home visit program was operating at capacity before the campaign, the District sought to streamline promotion and installation. CAFD developed an automated message recorded by the fire safety educator they delivered to residents using Carlton County's emergency warning system,

**Table 3.** Grand Rapids: how did the caller requesting smoke alarm installation report hearing about the smoke alarm program?*

| | October 2017 | November 2017 | December 2017 | January 2018 |
|---|---|---|---|---|
| *Automated call* | 12 | 19 | 10 | 5 |
| Banner | 27 | 16 | 11 | 1 |
| Billboard | 0 | 1 | 2 | 2 |
| Canvas | 0 | 5 | 0 | 4 |
| Event | 22 | 14 | 15 | 7 |
| Fire Dept. Personnel | 2 | 2 | 2 | 3 |
| Mailing | 21 | 33 | 36 | 41 |
| Media | 9 | 14 | 8 | 6 |
| Neighborhood Assoc. | 1 | 1 | 4 | 10 |
| Social media | 0 | 1 | 2 | 1 |
| Word of mouth | 32 | 26 | 36 | 41 |
| Other | 3 | 4 | 3 | 2 |
| Total | 129 | 136 | 129 | 123 |

Italicized text indicates the new promotion strategy implemented.
*Shaded columns indicate the months when automated calls were made.

which permitted calls to landline numbers enrolled to receive nonemergency alerts. A total of 2090 phone numbers in the Scanlon and Cloquet townships received calls over 10 days spread across 3 months (January–March 2018). CAFD staggered the calls to prevent exceeding volunteer capacity for home visits. In addition, postcards with information about the program were mailed to 319 households in the Brevator Township in October 2017, which recently contracted to join the CAFD's service area.

The automated calls went to geographically close households on a rolling basis which helped cluster home visits scheduled after each round of calls to a concentrated area. This resulted in reduced driving time between appointments and allowed CAFD volunteers to offer more home visit appointments and complete more installations. The automated calls yielded 28 smoke alarm installation requests for a 1% return of the calls; the postcards resulted in 20 requests for a home visit, a 6% return (Table 4).

# DISCUSSION

Automated calls can help increase demand for smoke alarms, or streamline installations, but fire departments should carefully consider their own programs to identify how dissemination strategies can best complement existing program efforts and department capacity to expand and/or streamline uptake. Across all sites, residents' uptake of smoke alarm installation services was between 0.02% and 2% of the calls placed, which is consistent with the 0.5% return documented in Baltimore City.[18] Therefore, fire departments need to be willing to place many more calls than requests they can fulfill in order to yield any increase in demand. All three participating sites staggered their automated calls over time, which allowed them to gauge the community response and plan subsequent call waves to

**Table 4.** Cloquet: how did the resident hear about the smoke alarm program?*

| | October 2017 | November 2017 | December 2017 | January 2018 | February 2018 | March 2018 |
|---|---|---|---|---|---|---|
| *Automated call* | 0 | *0* | *0* | *15* | *9* | *4* |
| *Postcards* | *14* | *4* | *2* | *0* | *0* | *0* |
| Fire Dept. Personnel | 2 | 6 | 2 | 1 | 0 | 0 |
| Flyers & Print Media | 7 | 7 | 6 | 4 | 4 | 4 |
| Social media | 0 | 0 | 1 | 1 | 0 | 0 |
| Word of mouth | 3 | 8 | 13 | 5 | 7 | 10 |
| Other | 1 | 0 | 2 | 3 | 0 | 1 |
| Total | 27 | 25 | 26 | 29 | 20 | 19 |

Italicized text indicates the new promotion strategy implemented.
*Shaded columns indicate the months when postcards were sent (October) and automated calls were made (January, February, and March).

better match department installation capacity. We note here that the participating fire departments included those serving urban and suburban populations as well as smaller rural communities. This demonstration of effect builds on previous work that focused on expanding uptake in one urban setting[18] and provides greater insight into how to effectively increase the presence of working smoke alarms in diverse community contexts.

An important lesson from this experience is for officials planning a smoke alarm installation campaign to identify and lean on potential resources beyond their own departments, and to communicate clearly with external partners about the strengths and weaknesses of the infrastructure used to implement the campaign. The three fire departments used different types of automated call systems (the Grand Rapids analog system had limited capacity to make calls at one time, for example) and different infrastructures for managing those systems (the Carlton County emergency warning system used by Cloquet offered geofencing and other features to help track where calls were made). Logistical considerations can affect the approach to disseminating smoke alarm campaign messages, but in the three participating sites logistics did not hinder their ability to use automated calls to promote their programs.

One logistical consideration that does limit the reach of the automated call approach to smoke alarm program dissemination is the current reliance of automated call systems on landlines. In all three participating sites, the automated call numbers were limited to landlines. This was also true of the Baltimore City system that informed this project.[18] With the increasing reliance on cellular phones, the percentage of households with landlines continues to decline nationally.[22] As such, there is a need to assess whether automated call dissemination that includes cellular phones is a viable and effective strategy for increasing smoke alarm installation program uptake.

These contributions to the literature and practice of fire prevention are the result of a process that does have some limitations. This is an observational study. The automated calls and dissemination strategies developed in response to local needs and resources and without standardization of the intervention or its implementation. Although each fire department used automated calls as part of their smoke alarm campaign, their differing approaches and resources does not allow for comparisons between the departments. Additional components (such as post card mailings) were included in some but not all sites, and this affects our confidence in attributing uptake solely to the automated call component of the intervention. For both of these limitations, we viewed the need to be responsive to local input as more important than imposing strict controls on the intervention components.

While the limitations are important to consider, the program findings reported here have several strengths. Most notably, this program is a partnership between an academic injury center and fire departments in three distinct communities. As such, the program development, implementation, and resulting impacts occurred in real world settings and reflect the adaptations needed to account for local resources and needs. By inviting local adaptation when testing different implementation strategies, this work revealed nuances in the process of promoting smoke alarm installation programs among communities. In addition, the automated call strategy builds directly on the findings from previous work, and addresses a neglected space in the literature, which is how to increase uptake of a known effective intervention (smoke alarms) and an effective implementation strategy (installation). The experiences of the three fire departments offer concrete guidance for those in the fire service and allied stakeholders for how to increase the prevalence of working smoke alarms in communities.

Smoke alarms are an effective strategy for reducing death when residential fires occur. Assuring that working smoke alarms are installed in homes can be accomplished through installation programs that offer free smoke alarms in response to community request, but those efforts to date have fallen short of complete coverage and people continue to die in fires in homes without working smoke alarms. Increasing uptake of smoke alarm installation programs can occur through fire department efforts to increase demand through automated calls.

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